2.4.1 Whole-system and individual-body analysis

Draw a free-body diagram for each object before choosing a whole-system or individual-body method. Bodies moving together along the same line with the same acceleration can be treated as one system. Internal forces cancel, so external force and total mass give acceleration. Then analyse one body to find tension or contact force.

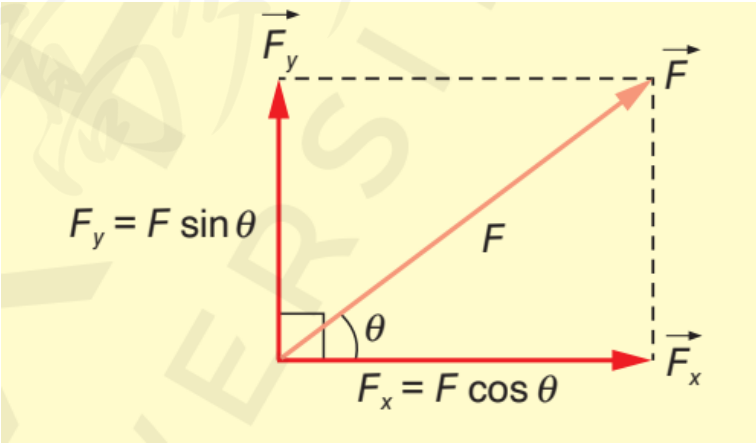
Bodies connected by an inextensible string over a pulley may have equal acceleration magnitudes in different directions. Choose an axis for each body, write its force equation and combine the equations using the string constraint. The same string has equal tension in the ideal light-string, frictionless-pulley model.

2.4.2 Resolving a force into perpendicular components

A force can be replaced by perpendicular components with the same combined effect. If its angle is measured from the positive x-axis, the adjacent component uses cosine and the opposite component uses sine. Measuring from the vertical interchanges these roles. Assign signs according to the chosen axes and each component's direction.

$$F_x = F\cos\theta$$
$$F_y = F\sin\theta$$

F: force (N); θ: angle measured from the positive x-axis to F. Components are signed.



θ is measured from the horizontal component. The adjacent component is F cos θ and the opposite component is F sin θ.